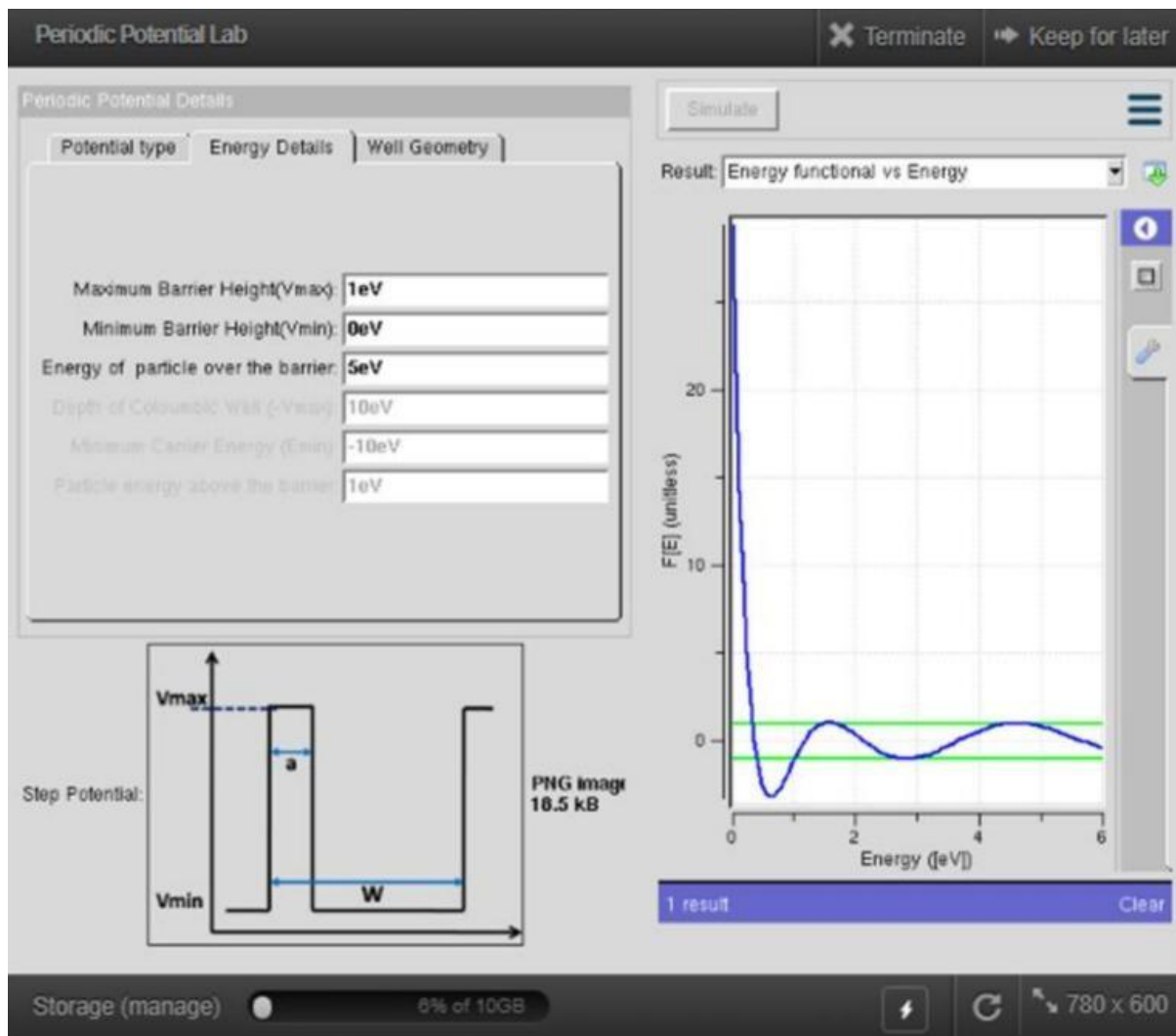
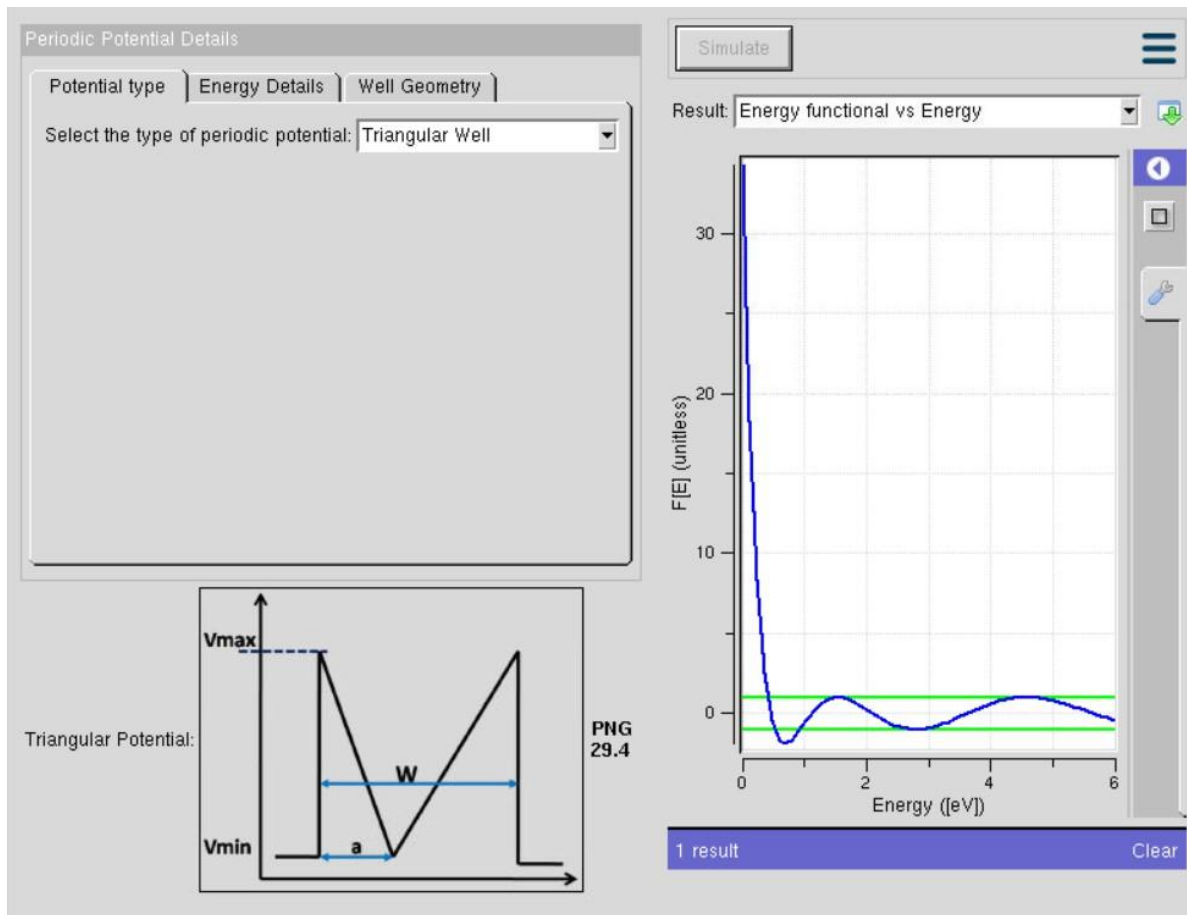


Exp- 4 Kronig Penney Model



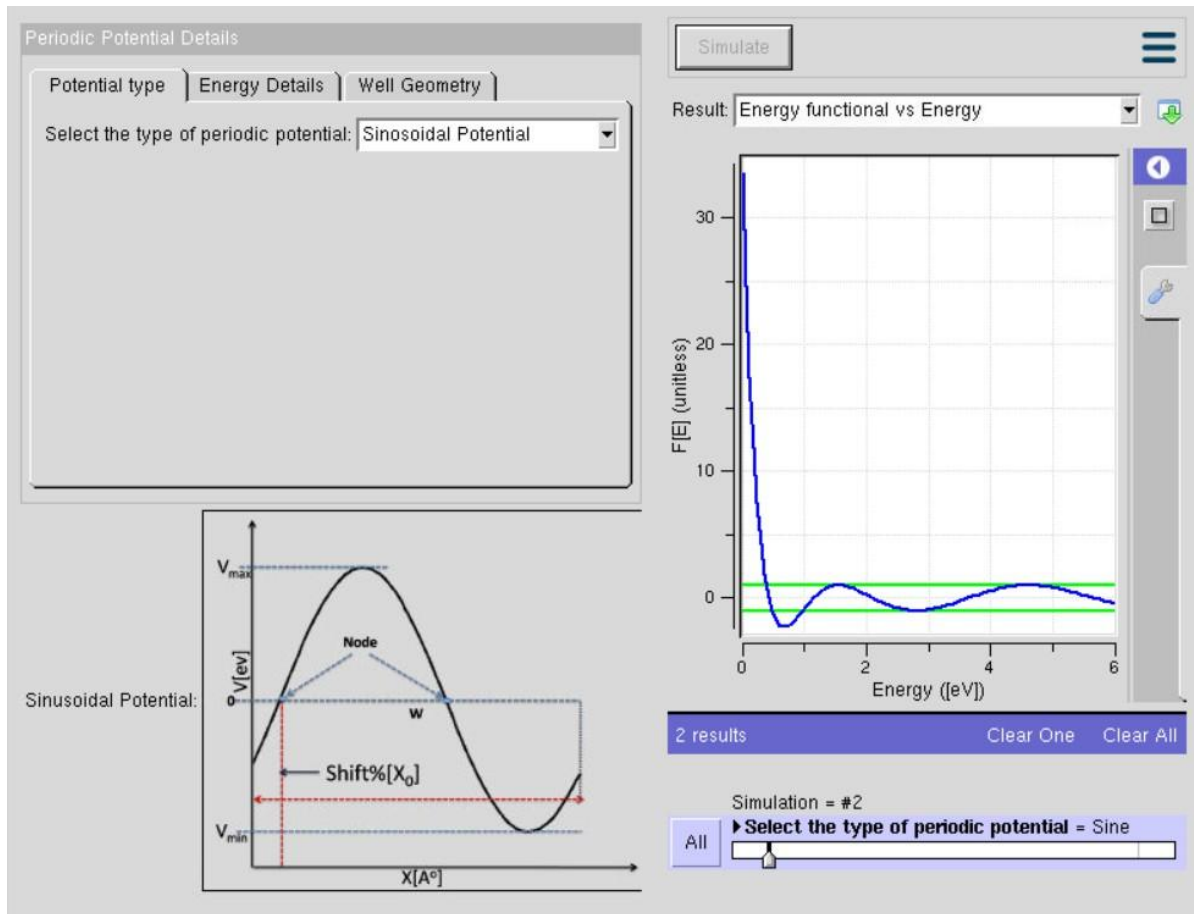
This image shows the Periodic Potential Lab simulation, which analyzes the movement of a particle through a periodic semiconductor potential. The simulation uses a barrier height of 1 eV and particle energy of 5 eV, allowing the particle to pass through the barriers. The graph of Energy Functional ($F(E)$) vs Energy illustrates the allowed energy states and electron behavior within the periodic structure. Overall, the results show how periodic potentials influence electron transport and energy band formation, helping researchers optimize semiconductor devices such as transistors, quantum wells, lasers, and other nanoelectronic components.

Case study-1: Simulate the output for Triangular well



This image shows the **Periodic Potential Lab** simulation with a **Triangular Well** potential. The triangular potential profile is displayed on the left, while the graph on the right shows the **Energy Functional ($F(E)$) versus Energy**, illustrating the allowed energy states of electrons within the periodic structure. The oscillating curve indicates how electron energy changes as it moves through the triangular potential wells. The simulation demonstrates that the shape of the potential well significantly affects electron confinement and energy band formation. Overall, this analysis is useful for designing and optimizing **semiconductor devices, quantum well structures, lasers, photodetectors, and other nanoelectronic applications** by improving carrier transport and device performance.

Case Study 2: Simulate the output for Sinusoidal Potential



This image shows the **Periodic Potential Lab** simulation using a **Sinusoidal Potential**. The sinusoidal potential profile on the left represents a smooth periodic variation in potential energy, similar to the periodic atomic arrangement in a crystal lattice. The graph on the right displays the **Energy Functional ($F(E)$) versus Energy**, showing how the electron energy varies within the sinusoidal potential. The oscillating curve indicates the allowed energy states and electron behavior in the periodic structure. Overall, this simulation demonstrates how a sinusoidal potential influences electron transport and energy band formation, making it valuable for designing **semiconductor devices, superlattices, quantum wells, photonic devices, and other nanoelectronic applications** with improved efficiency and performance.